

climatic data centres, hospitality data, and sensors, such massive datasets are also collected through social media. Businesses use data to gain insight into consumers' preferences, improve product design, and increase efficiency.

Big Data is here to stay, and companies and organisations that can effectively harness its power will have a competitive advantage in the coming years.

Hadoop and Big Data

A framework for distributed data analysis called Hadoop is available for free. It is designed for the storage, processing and analysis of large datasets in a distributed computing environment. Doug Cutting and Mike Cafarella created it in 2005 for processing large amounts of web data.

Hadoop is constructed on top of the several machines in a cluster to store data using the Hadoop Distributed File System (HDFS). Its key benefit is its extensibility. Hadoop also offers fault tolerance, as data is replicated over multiple nodes, which ensures that if a node fails, data can still be accessed.

Big Data focuses on:

- Velocity - How quickly data is generated, gathered and analysed
- Volume - The quantity of information generated
- Variety - The data type may be structured, informal, etc
- Veracity - The quality and precision of the data
- Value - How to use data to provide a snapshot of business processes

How HDFS works

HDFS is designed to store large files (petabytes and beyond) reliably and efficiently. In an HDFS cluster, large files are divided into smaller blocks and distributed among numerous machines. Every block is replicated on several machines to guarantee fault tolerance and data redundancy. HDFS is optimised for data streaming, rather than random access, making it well

suited for applications that involve large scale data processing, such as data warehousing and log processing.

In addition to storing and retrieving data, HDFS offers numerous tools to govern and track the data stored in a cluster, which includes data compression, encryption, and access control.

How MapReduce works underneath the Hadoop system

HDFS integrates MapReduce to offer a parallel processing architecture for distributed computing. MapReduce serves as a programming model and software framework that makes it possible to process large datasets in such environments.

The core idea of MapReduce is to split large datasets into smaller segments, process each segment separately on different nodes within the cluster, and consolidate the results into a final output. Every node executes a map function that converts the key-value pairs generated from input data. The outcomes of the map function are aggregated and reduced using a reduce function, which generates a final output. This technique enables the processing of vast amounts of data in a scalable and resilient fashion.

The Hadoop ecosystem

Hadoop is a framework for distributed data analysis, which is designed to enable processing and storage. It uses distributed computing to analyse large datasets.

The Hadoop ecosystem includes various components that work together to form a comprehensive Big Data platform. Here are some of its major components.

- The Hadoop Distributed File System or HDFS stores and manages immense quantities of data across an ensemble of Hadoop machines.
- MapReduce is the programming model implemented in Hadoop to process large datasets. It involves

dividing a large dataset into smaller chunks and independently processing each chunk in parallel across several nodes in a Hadoop cluster.

- YARN (Yet Another Resource Negotiator) is the cluster management technology used in Hadoop. Its primary function is to manage and allocate resources (such as CPU and memory) across the nodes in a Hadoop cluster to applications as needed.
- A data warehousing tool termed Hive delivers an SQL-like interface to manage and query data residing in Hadoop. Users can interact with tremendous data sets using typical SQL syntax because the algorithm converts queries generated by SQL into MapReduce workflows.
- Pig is a sophisticated programming framework for developing MapReduce scripts. Developers can get started with Hadoop effortlessly since it supplies an elementary scripting language that abstracts away a lot of the complexity involved in developing MapReduce programs.
- datasets is a powerful data processing engine capable of processing large data sets quickly and efficiently within Hadoop. It supports a wide variety of programming languages, notably Java, Python, and Scala, and is capable of analysing data in real-time or in batches.
- HBase is a non-relational database that is built on top of Hadoop. It proposes quick reading and writing access to data that's stored in Hadoop and provides optimal performance for preserving and analysing big amounts of organised information.
- ZooKeeper is a distributed coordination service utilised for managing and synchronising the configuration and state of services running in a Hadoop cluster.